

Relation Between Supergraph-Based Frequency (Freq_G) and Exact-Match Frequency ($\text{freq}_{\text{exact}}$)

1 Definitions

Let $G = (V, E)$ be a graph. Let $\mathcal{G}(G)$ denote the set of all connected subgraphs of G .

Definition 1 (Exact-match frequency). For a pattern P , the **exact-match frequency** counts the number of distinct connected subgraphs of G that are isomorphic to P :

$$\text{freq}_{\text{exact}}(P) = |\{H \in \mathcal{G}(G) : H \cong P\}|$$

Definition 2 (Supergraph-based frequency (proposed in the paper)). The **supergraph-based frequency** counts the number of connected subgraphs of G that *contain* an isomorphic copy of P :

$$\text{Freq}_G(P) = |\text{UP}_G(P)| = |\{H \in \mathcal{G}(G) : P \subsetneq H\}|$$

where $P \subsetneq H$ means $\exists H' \subseteq H$ such that $H' \cong P$ (subgraph isomorphism).

2 Exact Relation

Theorem 1 (Summation over the subgraph lattice). *For any pattern P and any graph G :*

$$\text{Freq}_G(P) = \sum_{\substack{Q \in \mathcal{G}(G)/\cong \\ P \subsetneq Q}} \text{freq}_{\text{exact}}(Q)$$

where $\mathcal{G}(G)/\cong$ is the set of isomorphism classes of connected subgraphs of G . Equivalently:

$$\text{Freq}_G(P) = \text{freq}_{\text{exact}}(P) + \sum_{\substack{Q \in \mathcal{G}(G)/\cong \\ P \subsetneq_i Q}} \text{freq}_{\text{exact}}(Q)$$

where $P \subsetneq_i Q$ means P is isomorphic to a proper subgraph of Q .

Proof. Every subgraph $H \in \mathcal{G}(G)$ belongs to exactly one isomorphism class $[H] \in \mathcal{G}(G)/\cong$. We

can therefore partition $UP_G(P)$ by the isomorphism class of its elements:

$$\begin{aligned}
\text{Freq}_G(P) &= |\{H \in \mathcal{G}(G) : P \subsetneq H\}| \\
&= \sum_{H \in \mathcal{G}(G)} \mathbf{1}[P \subsetneq H] \\
&= \sum_{Q \in \mathcal{G}(G)/\cong} \sum_{\substack{H \in \mathcal{G}(G) \\ H \cong Q}} \mathbf{1}[P \subsetneq H] \\
&= \sum_{Q \in \mathcal{G}(G)/\cong} \mathbf{1}[P \subsetneq Q] \cdot |\{H \in \mathcal{G}(G) : H \cong Q\}| \\
&= \sum_{\substack{Q \in \mathcal{G}(G)/\cong \\ P \subsetneq Q}} \text{freq}_{\text{exact}}(Q).
\end{aligned}$$

The key step (line 3 to 4) uses the fact that $P \subsetneq H$ depends only on the isomorphism class of H : if $H \cong Q$ and $P \subsetneq Q$, then $P \subsetneq H$.

Separating the term $Q = P$ (where $P \subsetneq P$ trivially via the identity) gives the second form. $\square$

3 Interpretation

3.1 Freq_G as a cumulative sum over the subgraph lattice

The set of isomorphism classes $\mathcal{G}(G)/\cong$ forms a partially ordered set (poset) under $\subsetneq$. The function $\text{freq}_{\text{exact}}$ assigns a count to each element of this poset, and Freq_G is its **cumulative sum upward** (the zeta transform):

$$\text{Freq}_G(P) = (\zeta \cdot \text{freq}_{\text{exact}})(P) = \sum_{Q \geq P} \text{freq}_{\text{exact}}(Q)$$

where $Q \geq P$ means $P \subsetneq Q$ in the poset ordering. This is a standard construction in combinatorics on partially ordered sets.

3.2 Recovering $\text{freq}_{\text{exact}}$ from Freq_G (Möbius inversion)

By Möbius inversion on the poset $(\mathcal{G}(G)/\cong, \subsetneq)$, the exact frequency can be recovered:

$$\text{freq}_{\text{exact}}(P) = \sum_{\substack{Q \in \mathcal{G}(G)/\cong \\ P \subsetneq Q}} \mu(P, Q) \cdot \text{Freq}_G(Q)$$

where μ is the Möbius function of the subgraph poset, defined recursively by $\mu(P, P) = 1$ and

$$\mu(P, Q) = - \sum_{\substack{R \\ P \subsetneq R \subsetneq_i Q}} \mu(P, R) \quad \text{for } P \subsetneq_i Q.$$

Computing μ is intractable in general (the subgraph lattice is exponentially large), but this shows that Freq_G and $\text{freq}_{\text{exact}}$ are *in bijection*: knowing one determines the other completely.

3.3 Why Freq_G is the right measure for anomaly detection

The relation $\text{Freq}_G(P) = \text{freq}_{\text{exact}}(P) + \sum_{Q \supsetneq P} \text{freq}_{\text{exact}}(Q)$ means:

- $\text{freq}_{\text{exact}}(P)$ counts the occurrences of P *alone* — how many times P appears as an exact copy.
- $\text{Freq}_G(P)$ counts the occurrences of P *in any context* — how many subgraphs of G contain P as part of their structure, regardless of what else they contain.

For anomaly detection, Freq_G is more appropriate because:

1. A pattern P that is rare as an exact match but common inside larger structures is *not* anomalous — it is a building block of normal behavior. Only $\text{Freq}_G(P)$ captures this: if many larger subgraphs contain P , then $\text{Freq}_G(P)$ is high, correctly indicating normality.
2. Conversely, if $\text{Freq}_G(P)$ is low, then P is rare both as an exact match *and* as a substructure of any larger pattern. This is a stronger (and more useful) statement of anomalousness.

4 Bounds

Proposition 2 (Bounds on Freq_G in terms of $\text{freq}_{\text{exact}}$). *For any pattern P :*

$$\text{freq}_{\text{exact}}(P) \leq \text{Freq}_G(P) \leq \sum_{s=|V(P)|}^n N_s(G)$$

where $N_s(G) = |\{H \in \mathcal{G}(G) : |V(H)| = s\}|$ is the number of connected subgraphs of G with exactly s nodes.

The lower bound is tight when P is never contained in any strictly larger connected subgraph of G (e.g., $P = G$ itself). The upper bound is tight when P is a single node or edge (contained in every connected subgraph of sufficient size).

Proposition 3 (Anti-monotonicity from the summation identity). *If $P_1 \subseteq_i P_2$, then $\{Q : P_2 \subseteq Q\} \subseteq \{Q : P_1 \subseteq Q\}$ by transitivity of $\subseteq$. Therefore:*

$$\text{Freq}_G(P_2) = \sum_{Q \supseteq P_2} \text{freq}_{\text{exact}}(Q) \leq \sum_{Q \supseteq P_1} \text{freq}_{\text{exact}}(Q) = \text{Freq}_G(P_1)$$

since the sum for P_2 runs over a subset of the terms in the sum for P_1 , and all terms $\text{freq}_{\text{exact}}(Q) \geq 0$.

5 Examples

Example 1 (Single edge). Let $P = K_2$ (a single edge). Then $P \subseteq H$ for every connected subgraph H with $|V(H)| \geq 2$. So:

$$\text{Freq}_G(K_2) = \sum_{\substack{Q \\ |V(Q)| \geq 2}} \text{freq}_{\text{exact}}(Q) = |\mathcal{G}(G)| - n$$

since the only connected subgraphs NOT containing an edge are the n isolated nodes. Meanwhile $\text{freq}_{\text{exact}}(K_2) = |E|$.

Example 2 (Triangle). Let $P = K_3$ (a triangle). Then:

$$\text{Freq}_G(K_3) = \text{freq}_{\text{exact}}(K_3) + \sum_{Q \supsetneq K_3} \text{freq}_{\text{exact}}(Q)$$

The first term counts the triangles in G . The second counts every larger connected subgraph of G that contains at least one triangle. In dense graphs, $\text{Freq}_G(K_3) \gg \text{freq}_{\text{exact}}(K_3)$.

Example 3 (Anomalous clique). Let $P = K_7$ (a 7-clique injected as an anomaly into a sparse graph like Cora with average degree 4.1). Then:

- $\text{freq}_{\text{exact}}(K_7)$: the number of 7-cliques in G . In Cora with 10 injected cliques, $\text{freq}_{\text{exact}}(K_7) = 10$.
- $\text{Freq}_G(K_7) = 10 + \sum_{Q \supsetneq K_7} \text{freq}_{\text{exact}}(Q)$. Since Cora is sparse, there are very few connected subgraphs larger than 7 nodes that still contain a complete 7-clique. So $\text{Freq}_G(K_7) \approx \text{freq}_{\text{exact}}(K_7) = 10$. The two measures nearly coincide for large, rare patterns.

This illustrates that $\text{Freq}_G(P) \approx \text{freq}_{\text{exact}}(P)$ when P is large and rare — precisely the regime where anomaly detection operates.

6 Summary

The exact, deterministic relation between the two frequency measures is:

$$\text{Freq}_G(P) = \sum_{\substack{Q \in \mathcal{G}(G)/\cong \\ P \subsetneq Q}} \text{freq}_{\text{exact}}(Q) = \text{freq}_{\text{exact}}(P) + \sum_{Q \supsetneq P} \text{freq}_{\text{exact}}(Q)$$

Freq_G is the zeta transform of $\text{freq}_{\text{exact}}$ over the subgraph lattice, and $\text{freq}_{\text{exact}}$ is recoverable via Möbius inversion. For anomaly detection, where the patterns of interest are large and rare, the surplus $\sum_{Q \supsetneq P} \text{freq}_{\text{exact}}(Q)$ is small, making $\text{Freq}_G(P) \approx \text{freq}_{\text{exact}}(P)$. For small, common patterns, the surplus is large, but this is desirable: it correctly prevents flagging common building blocks as anomalous.